③ Clock Sync. and impact of Clock offset on Distributed sensor data

Intro: Clocks not ideal

- granularity (g How often clock updates)
- Drift (deviate from realtime)

$$\underbrace{\left|\frac{dC_k(t)}{dt} - 1\right|}_{\text{rate of local clock}} > \rho_k \leftarrow \text{Tolerance (Max drift rate)}$$

- often Modelled as Cont. differentiable functions $C_k(t)$

Precision: $|C_i(t) - C_k(t)| \leq \pi$

how close two local clocks are

Accuracy: $|C_k(t) - t| \leq \alpha$

How close local clock is to ideal ref clock (f× UTC)

Clock Sync: (because drift)

- such as Network Time Protocol (NTP)

Goal in distributed: all clocks agree ie. Minimize Precision of Clocks